UNIFORM D MAKING THE TWIST MACHINERY WORK ON A QUASI-COMPACT OPEN

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Where invertible. lemma-where-invertible (carded this batch). Parts (2), (3), (4) follow from Y contained in W_d.
- Trivialization on charts. lemma-when-invertible (carded this batch). Alternative characterization (1) route for the containment.
- Principal opens of W_d. lemma-principal-open (carded this batch). Part (5): W_s = D_+(f) intersect W.

Source reference. Original equation 01MP: sources/stacks-constructions/pinned-original-source.tex, source lines 1949–1949. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 01MO: sources/stacks-constructions/pinned-original-source.tex, source lines 1930–1930. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 01MQ: sources/stacks-constructions/pinned-original-source.tex, source lines 1966–1966. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01MV: sources/stacks-constructions/pinned-original-source.tex, source lines 2136–2136. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01MS: sources/stacks-constructions/pinned-original-source.tex, source lines 1986–1986. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01MU: sources/stacks-constructions/pinned-original-source.tex, source lines 2070–2070. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a graded ring. Let $X = Proj(S)$. Let $Y \subset X$ be a quasi-compact open subscheme. Denote $\mathcal{O}_Y(n)$ the restriction of $\mathcal{O}_X(n)$ to Y. There exists an integer $d \geq 1$ such that*

1. *the subscheme Y is contained in the open W_d defined in Lemma 01MU,*
2. *the sheaf $\mathcal{O}_Y(dn)$ is invertible for all $n \in \mathbf{Z}$,*
3. *all the maps $\mathcal{O}_Y(nd) \otimes_{\mathcal{O}_Y} \mathcal{O}_Y(m) \longrightarrow \mathcal{O}_Y(nd + m)$ of Equation (01MO) are isomorphisms,*
4. *all the maps $\widetilde{M}(nd)|_Y = \widetilde{M}|_Y \otimes_{\mathcal{O}_Y} \mathcal{O}_X(nd)|_Y \to \widetilde{M(nd)}|_Y$ (see 01MQ) are isomorphisms,*
5. *given $f \in S_{nd}$ denote $s \in \Gamma(Y, \mathcal{O}_Y(nd))$ the image of f via (01MP) restricted to Y, then $D_+(f) \cap Y = Y_s$,*
6. *a basis for the topology on Y is given by the collection of opens Y_s, where $s \in \Gamma(Y, \mathcal{O}_Y(nd))$, $n \geq 1$, and*
7. *a basis for the topology of Y is given by those opens $Y_s \subset Y$, for $s \in \Gamma(Y, \mathcal{O}_Y(nd))$, $n \geq 1$ which are affine.*

Proof. Since Y is quasi-compact there exist finitely many homogeneous $f_i \in S_+$, $i = 1, \ldots, n$ such that the standard opens $D_+(f_i)$ give an open covering of Y. Let $d_i = \deg(f_i)$ and set $d = d_1 \ldots d_n$. Note that $D_+(f_i) = D_+(f_i^{d/d_i})$ and hence we see immediately that $Y \subset W_d$, by characterization (2) in Lemma 01MU or by (1) using Lemma 01MS. Note that (1) implies (2), (3) and (4) by Lemma 01MU. (Note that (3) is a special case of (4).) Assertion (5) follows from Lemma 01MV. Assertions (6) and (7) follow because the open subsets $D_+(f)$ form a basis for the topology of X and are affine. □